Project Report

Abstract

The AI Job Assistant is a student-focused job-prep platform that streamlines the entire path from drafting a resume to becoming interview-ready. Built as a modular Python system with a Gradio UI, it handles document ingestion and PDF/DOCX text extraction, then parses resumes into structured Pydantic models using a local Ollama (Qwen3 or Qwen3.5) LLM with an optional Gemini fallback.

For candidate-facing features it offers resume-grounded RAG chat (MiniLM embeddings + in-memory Qdrant), ATS-style scoring of resumes against job descriptions via LLM evaluation (with a heuristic fallback), gauge/radar visualizations, missing-keyword analysis, and clear, actionable suggestions. Job search is integrated via the RemoteOK feed with client-side filtering and semantic ranking (cosine similarity) to prioritize matches.

A dashboard tracks applications and analytics in one place, while a voice-based mock-interview engine—using Faster Whisper, Qwen ASR, or Wav2Vec2 for ASR plus lightweight TTS and a local LLM interview flow—delivers interactive practice and post-interview performance reports. The design keeps student workflows consistent and efficient and supports privacy-conscious local execution where possible.

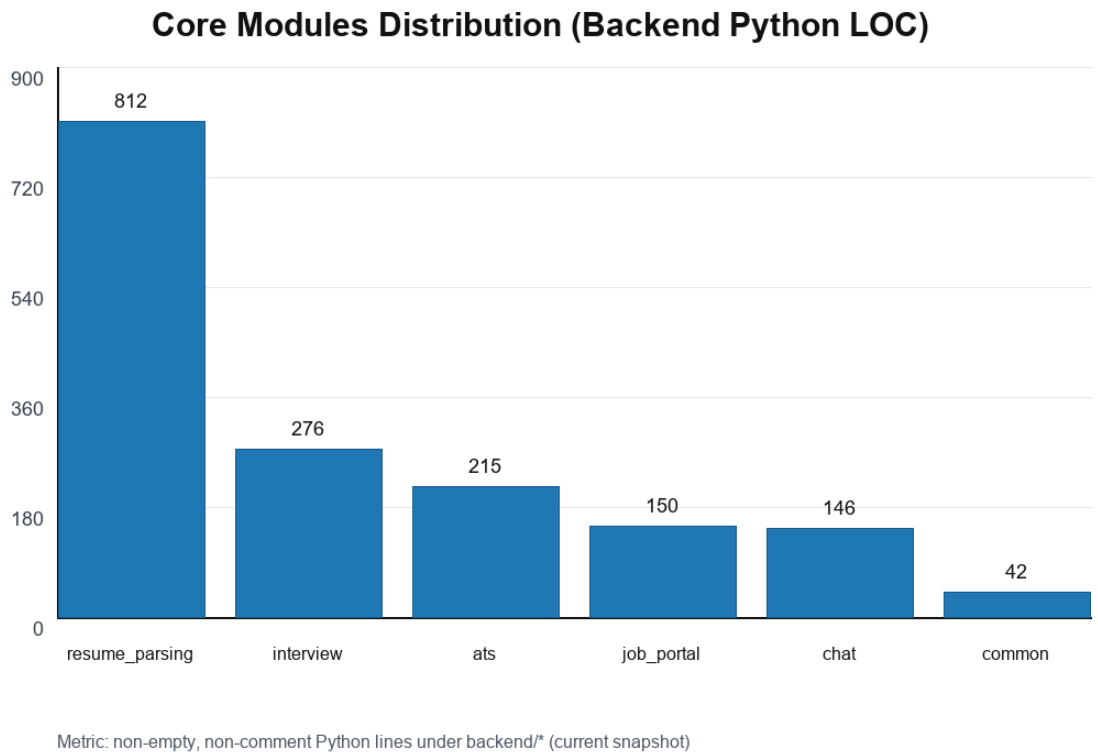

Fig 1: Core Modules Distributions

Chapter 1

Introduction

1.1 Background

In today's digital hiring landscape, students preparing for internships and entry-level roles must manage multiple steps such as building a resume, validating ATS compatibility, searching for suitable jobs, tracking applications, and practicing interviews. However, most learners rely on separate tools for each stage, including resume templates, keyword checkers, job portals, trackers, and interview practice platforms. This fragmented approach increases time spent switching contexts, produces inconsistent feedback, and makes it difficult to maintain a clear improvement loop from resume refinement to interview readiness.

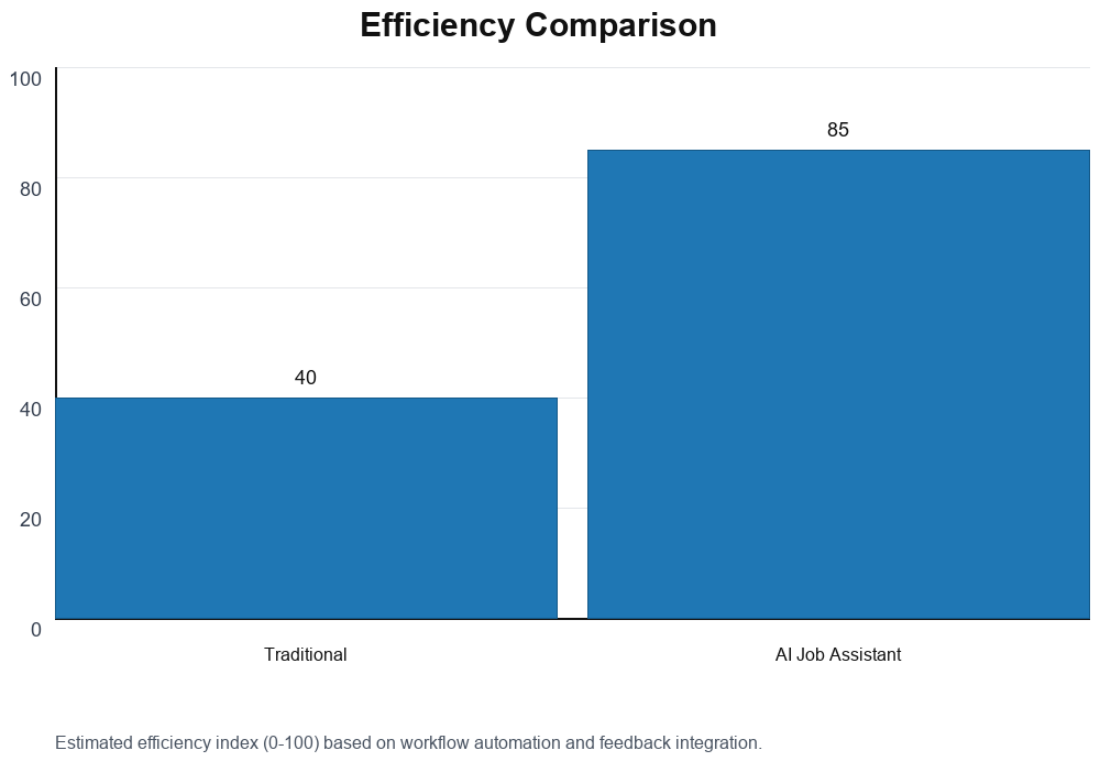

Fig 2: Efficiency Comparison

1.2 Problem Statement

The primary problem addressed in this project is the absence of a unified, student-oriented job preparation platform that can connect resume understanding, ATS readiness, job discovery, application tracking, and interview practice into one guided workflow. Existing approaches typically suffer from:

- Manual and repetitive effort to tailor resumes for different job descriptions
- Limited, inconsistent feedback on ATS keyword gaps and overall profile quality
- Inefficient job discovery due to basic keyword search and poor personalization
- Lack of structured, interactive interview practice with actionable end-of-session feedback
- Difficulty tracking progress across resume, job matching, applications, and interview performance

These challenges become more significant during placement seasons, where students must iterate quickly and improve application quality under time constraints.

1.3 Objective of the Project

The main objective of this project is to develop a comprehensive, student-oriented AI Job Assistant platform that:

- Extracts and structures resume information from PDF/DOCX into a consistent candidate profile
- Evaluates ATS compatibility against job descriptions and provides keyword-gap analysis with actionable improvement suggestions
- Enables interactive resume-based Q&A through retrieval-augmented chat for quick self-review and clarification
- Supports job discovery and shortlisting by fetching remote opportunities and ranking them using semantic similarity to the candidate profile
- Helps students track applications and monitor progress through a dashboard with analytics
- Provides mock interview practice with voice interaction and generates an end-of-session performance report
- Encourages faster iteration and better decision-making by presenting clear visual feedback through charts and summaries

1.4 Scope of the System

The system is designed to support students and early-career candidates preparing for internships and entry-level roles. It offers an integrated workflow covering resume parsing, ATS scoring, resume chat (RAG), job search and semantic matching, application tracking dashboard analytics, and mock interview practice within a single interface. The platform is intended to run locally where possible, for example with Ollama for LLM inference and local embeddings for retrieval, while still allowing optional cloud fallback for extraction and scoring when configured. Its scope is to improve job-preparation efficiency, visibility, and feedback quality rather than automate the entire hiring process.

Chapter 2

Literature Survey

2.1 Existing Systems

Existing student job-preparation solutions are often fragmented and require learners to use multiple independent tools for different stages of the hiring process. In practice, students may use one tool for resume formatting, another for ATS keyword checking, separate job portals for searching roles, a spreadsheet for application tracking, and a different platform for interview practice. This separation reduces continuity across the preparation workflow and makes it difficult to maintain a consistent feedback loop from resume edits to application progress and interview performance.

Some key limitations of common existing systems include:

- Fragmentation: Lack of integration between resume building, ATS scoring, job discovery, application tracking, and interview practice
- Limited feedback depth: Basic keyword counts without structured explanations, category-level breakdowns, or prioritized suggestions
- Manual processes: Heavy reliance on manual resume tailoring, repetitive copying and pasting of job descriptions, and manual tracking of progress
- Restricted personalization: Job matching often relies on keywords rather than semantic similarity to the candidate profile
- Poor practice-to-improvement loop: Interview preparation tools may not connect back to resume strengths, weaknesses, and role requirements
- Privacy constraints: Cloud-only tools may require uploading personal documents and audio, which some students may prefer to avoid
- Basic reporting: Limited analytics that summarize progress across ATS readiness, job matching, application status, and interview performance

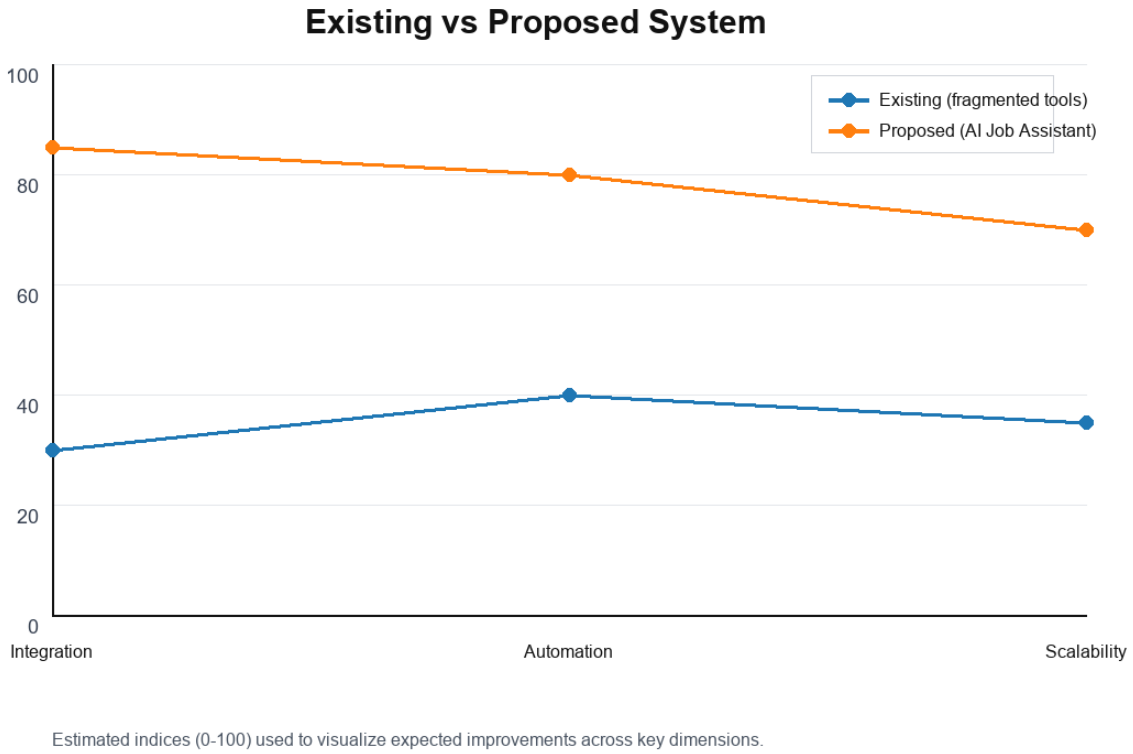

Fig 2.1: Existing vs Proposed System

Existing Approach / System	Strength	Limitations
Standalone resume template/builders	Helps students format resumes quickly	Does not validate ATS compatibility against specific job descriptions; limited structured feedback and role targeting
ATS keyword checker tools	Highlights missing keywords for a JD	Often shallow (keyword-only), not connected to the student's full profile; weak explanation of trade-offs and prioritization
Job portals and basic search (keyword filters)	Large set of job listings and quick discovery	Poor personalization; results depend heavily on keywords and manual filtering; no semantic match to resume content
General-purpose chatbots	Fast Q&A and drafting support	Not grounded in the student's resume unless manually pasted; prone to generic advice and inconsistent outputs
Interview question banks / mock tests	Provides practice questions and common patterns	Limited interactivity; typically not voice-based; weak linkage between resume/JD and the interview plan
Separate speech-to-text and note-taking tools	Captures spoken answers for review	Requires manual setup and manual evaluation; no integrated scoring or structured interview feedback
Manual tracking (spreadsheets/notes)	Flexible and customizable	Time-consuming, error-prone, and not real-time; hard to maintain a consistent improvement loop across resume, jobs, and interviews

Table 2.1: Literature Review

2.2 Proposed System

The proposed system introduces a unified, student-oriented AI Job Assistant that integrates multiple job-preparation functions into a single application. Instead of switching between disconnected tools, the platform provides a continuous workflow from resume parsing to ATS readiness, job discovery, application tracking, and interview practice.

Key features of the proposed system include:

- Unified workflow: One Gradio-based interface for resume parsing, ATS scoring, RAG chat, job search and matching, application tracking, and mock interviews
- AI resume understanding: PDF/DOCX text extraction followed by LLM-based structured extraction into a consistent candidate profile
- ATS scoring with feedback: Job-description evaluation using an LLM scorer with a heuristic fallback, including keyword gaps, suggestions, and visual score breakdowns
- Resume RAG chat: Retrieval-augmented Q&A grounded in the parsed resume using embeddings and an in-memory vector store

- Job discovery and matching: Remote job search combined with semantic similarity ranking to prioritize relevant opportunities
- Dashboard analytics: Application statistics, status tracking, and visual summaries to help students monitor their job-search pipeline
- Voice mock interviews: Interactive interview practice with speech-to-text options and text-to-speech prompts, plus an end-of-session report
- Local-first option: Supports local execution to reduce reliance on cloud services and improve privacy, with optional Gemini fallback when configured

2.3 Research Gap

From the study of existing systems, it is evident that there is still a lack of a fully integrated, student-oriented job preparation solution that combines resume understanding, ATS readiness, job discovery, application tracking, dashboard visibility, and interview practice in a single workflow. Most current tools address only one part of the process, which results in fragmented feedback, repeated manual work, and poor continuity from resume improvement to application outcomes.

The proposed system addresses this gap by:

- Providing a unified platform that connects resume parsing, ATS scoring, resume-based chat, job search and matching, application tracking, and mock interviews
- Enabling faster, more consistent feedback loops through structured scoring, keyword-gap analysis, dashboard analytics, and actionable suggestions
- Improving personalization of job shortlisting using semantic similarity rather than relying only on keyword filters
- Supporting interactive interview preparation, including voice-based practice, with end-of-session reporting to guide targeted improvement
- Offering a local-first execution path to reduce privacy concerns when handling personal documents and audio, with optional cloud fallback when configured
- Remaining modular and extensible so additional models, data sources, and evaluation components can be integrated as requirements evolve